TIKO — Testing Plan

Oct 10, 2026 · @ki

1. Objectives, scope and approach

This plan proves that TIKO meets its 93 functional requirements (48 P0, 34 P1, 11 P2) and the 17 non-functional targets before each release gate, with most checks automated because one developer builds and tests the product. It follows the Technical Specification, Database Design and API Documentation; the release criteria it must satisfy come from the PRD.

What testing must protect. The product promises are speed, no lost logs, kindness and privacy, so the plan puts its heaviest effort there:

- A tap shows in 100 ms and is saved in 300 ms; Today is interactive in under 2 seconds on a mid-range phone over 4G (NFR-01, NFR-02).
- No log is lost offline, on tab close or on a failed sync (NFR-04).
- Streaks, goals and the safe-to-spend number are right every time and identical on device and server (FR-ENG-01, FR-MON-06).
- No shaming language, no failed state, no diagnosis, a visible self-care disclaimer (NFR-17, FR-GOL-08, FR-MND-14).
- Journal text and exact amounts never reach analytics, notifications or logs (NFR-06).

Scope

In scope	Out of scope
Responsive PWA, guest mode and sign-in, habits, goals and treats, money, mind, reminders, offline and sync, settings and data rights	Native apps and the V2 wrapper
API routes, scheduled jobs, database constraints, shared engine	Automatic step and health-app import (V1)
P0 requirements in full; P1 requirements chosen for the MVP launch	Bank linking and any movement of money (must be proven absent)

In scope	Out of scope
Billing, savings goals and weekly recap are planned here and tested when built in V1	Social features, leaderboards (must be proven absent)

Approach

1. **Acceptance tests first.** The Functional Requirements ask for an acceptance test for every P0 requirement before development starts. Sections 3 and 4 list all 48.
2. **Test where the logic lives.** Rules sit in a pure engine package, so most correctness checks are fast unit and property tests, not browser tests.
3. **Test the failure paths.** Offline, retries, clock skew and conflicting edits get their own scenarios (section 6), because they are where lost-data bugs hide.
4. **Automate everything that can run on every change;** reserve people for what machines cannot judge: real devices, screen readers, tone of copy, and privacy review.
5. **Measure, do not assume.** Performance and accessibility targets are enforced as budgets in CI and confirmed on a real phone before beta.

2. Test levels, tools and environments

Six levels, from fastest to slowest; the lower the level, the more tests it holds and the earlier it runs.

Level	Code	What it checks	Tools	Runs
Unit and property	U	Engine rules (streaks, goals, budget, XP), validators, copy lint, formatters	Vitest, fast-check	Every commit
Integration	I	Sync routes, auth, rate limits, constraints, row-level security, jobs against a real Postgres	Vitest, a Postgres container, Mailpit for email	Every pull request
End-to-end	E	Critical flows in a real browser, including offline	Playwright	Pull request (smoke); nightly (full)

Level	Code	What it checks	Tools	Runs
Non-functional	N	Speed, accessibility, load, compatibility	Lighthouse CI, axe, k6, Playwright projects	Pull request (budgets); nightly and pre-release
Security and privacy	S	Scans and targeted attack tests, data audit	OWASP ZAP baseline, dependency and secret scanning, manual review	Nightly (scans); pre-release (review)
Manual	M	Real devices, screen readers, copy tone, install and push on phones	Checklists	Pre-beta and pre-release

Environments

Environment	Used for
Local and CI	Levels U, I, E with a throwaway database, a fake clock and stubbed email and push
Preview (per pull request)	Smoke E2E, Lighthouse and axe against a deployed build
Staging	Full E2E, load test, security scan, real-device and manual checks, beta users
Production	Smoke test after each release, uptime monitor; no destructive tests

Test data and controls

- **Fixed clock.** Anything that depends on the date (streaks, deadlines, budgets, quiet hours, magic-link expiry) runs on a controllable clock, never the real one.
- **Time zones.** Every date rule is run in Asia/Kolkata, UTC, America/Los_Angeles and a far-east zone such as Pacific/Kiritimati (UTC+14), including the midnight boundary and a mid-streak zone change.
- **Accounts.** Factories create users with no history, 7, 30 and 90 days of history, a full budget month, and a guest dataset; a second account exists in every security test.

- **Services.** Email goes to a local inbox, web push to a stub with controllable failures (including 410), Google sign-in uses a test account on staging, and Razorpay test mode is used when billing is built in V1.
- **No real personal data** is used in any environment; journal and mood fixtures are invented text.

3. P0 acceptance tests: account, habits, goals, treats

Each row is the check that proves one P0 requirement. Level codes are from section 2 (U unit, I integration, E end-to-end, M manual). A requirement passes only when every listed check passes.

ID	Check that proves it	Level
FR-ACC-01	"Try without signing up" starts in one tap; no account request is made; data survives a reload and stays on the device	E
FR-ACC-02	Magic link arrives in the test inbox and signs in; Google sign-in works on staging with a test account	E, M
FR-ACC-03	A guest with 3 habits, 20 logs and 1 goal signs in to a new account: all present, no duplicates. Into an account that already has "water": one water habit, logs combined. Calling merge twice changes nothing	I, E
FR-ACC-04	With a fake clock, a link works at 14:59 and fails at 15:01; a second use fails; the database holds only a hash	I
FR-ACC-05	Picking 3 habits, 1 goal with a treat, a reminder time and tone takes at most 5 screens and Today opens with the first card ready	E
FR-ACC-06	No permission prompt on first load; the prompt appears only after a reminder time is set (spy on the permission call)	E
FR-HAB-01	Each of the four modes logs correctly: tap counter, check-off, timer, manual number	U, E

ID	Check that proves it	Level
FR-HAB-02	A habit needs name, icon, mode, unit, target above 0, type and cadence; missing or invalid fields are rejected	U, I
FR-HAB-03	A tap updates the card in under 100 ms and stores the log; undo works at 9 seconds and is gone after 10	E, N
FR-HAB-04	The 7 core habits are available with default targets; activating one creates a row with its core key	U, E
FR-HAB-05	Create, rename, archive and reorder work; an archived habit keeps its history	E
FR-GOL-01	The engine supports all six goal types; the MVP screens offer five, since savings goals are V1 (see open question 1)	U
FR-GOL-02	At least 10 templates exist; a goal is created in 3 taps from a template, or from what, how much and by when	U, E
FR-GOL-03	Creating a goal past the cap is refused with a plain explanation; the cap holds on the server too	I, E
FR-GOL-04	After any log, edit, undo or delete, every active goal on that habit or category is recomputed and others are untouched	U, I
FR-GOL-05	A progress ring shows on Today and on the goal page and matches the stored progress	E
FR-GOL-06	One nudge at 50% and one at 90%; neither repeats after more logs or a reload	U, E
FR-GOL-07	The log that meets the target marks the goal complete and shows the celebration at once, online and offline	E
FR-GOL-08	A passed deadline sets status to due and offers extend, shrink or restart with history kept; the schema has no failed status, and no screen shows one	U, E

ID	Check that proves it	Level
FR-TRT-01	A treat stores name, optional photo, optional cost and free or paid; invalid values are rejected	U, I
FR-TRT-02	30 starter treats are listed by cost; a custom treat can be written	U, E
FR-TRT-03	A goal works without a treat and a treat can be added later	E
FR-TRT-04	The celebration shows the treat with two actions: Claim and Later	E
FR-TRT-05	Claim stores the date and puts the treat on the Trophy shelf; Later keeps it on the Goals tab	E

4. P0 acceptance tests: engagement, money, mind, notifications, offline

ID	Check that proves it	Level
FR-ENG-01	The streak cases in section 5 all give the expected result in every test time zone	U
FR-ENG-08	No leaderboard, rank or negative score exists on any screen or route; checked by a content review and a search of routes and copy	M, E
FR-MON-01	An expense is saved with an amount and one category in two taps; the date defaults to today	E
FR-MON-02	The 8 default categories exist; a custom category can be added	E
FR-MON-03	320.50 rupees is stored as 32050 and shown in the user's currency; no float or decimal column holds money	U, I
FR-MON-04	After saving, Undo is offered for 10 seconds and removes the expense; it is gone afterwards	E

ID	Check that proves it	Level
FR-MON-05	One monthly budget can be set, plus an optional limit per category	E, I
FR-MON-06	The safe-to-spend cases in section 5 give the expected value, never below zero, and update after each entry	U, E
FR-MON-13	No route, screen, dependency or permission links to a bank or moves money; confirmed by a code and dependency review	M
FR-MND-01	A mood on the 5-point face scale plus an energy level is saved in two taps and in under 10 seconds	E
FR-MND-02	Up to 3 tags and a one-line note are accepted; a fourth tag and a line break are rejected	U, E
FR-MND-04	A journal entry auto-saves while typing; the database column holds only ciphertext; decrypting returns the original text	I
FR-MND-05	Another user cannot read a journal entry; it is absent from analytics events, notifications and error reports; it is present in the owner's export	I, S
FR-MND-13	A mood of 1 and a mood of 5 earn the same reward; nothing is given for a particular mood	U
FR-MND-14	The self-care statement (not therapy, not medical advice) appears in onboarding and in Settings	E, M
FR-NOT-01	A daily reminder time and a tone (gentle, playful, minimal) can be saved and are used	E, I
FR-NOT-02	With a fake clock, never more than 2 notifications in a day and none inside quiet hours (default 10 pm to 8 am), in every test time zone	I
FR-NOT-03	Push is sent where permitted and email otherwise; each channel can be turned off; a 410 from the push service switches that user to email	I, M

ID	Check that proves it	Level
FR-NOT-06	Every tone's copy is free of words such as failed, lazy and behind and never contains journal text or exact amounts; a lint test scans the whole library	U
FR-SYN-01	The app can be installed to the home screen and opens with no connection after the first visit	E, M
FR-SYN-02	Habits, expenses and mood can be logged offline and are stored on the device first	E
FR-SYN-03	After reconnecting, queued entries sync by themselves within a minute; "Saved on this device" shows until the queue is empty	E
FR-SYN-04	The same entry edited on two devices keeps the later edit	I, E
FR-SYN-06	A failed sync (server error, timeout, oversized batch, network cut mid-request) retries and never deletes anything on the device	I, E

5. Engine test cases

The engine is the most important code to get right, so its cases are written out with expected values. Each runs as a unit test in every test time zone, and the shared ones also run against the server copy of the engine to prove both sides agree.

Streaks (FR-ENG-01)

Days are listed in order: M met, X missed, P paused. A streak counts met days. A forgiven day keeps the chain and shows a half-filled dot but adds nothing to the count (assumed). The rule tested is: a miss is forgiven when it is not right after another miss and no other day was forgiven in the previous 6 days.

#	Case	Days	Expected
S1	Plain run	M M M M	4
S2	One miss forgiven	M M X M M	4; day 3 is a half dot

#	Case	Days	Expected
S3	Two misses in a row	M M X X M	Chain breaks on day 4; streak is 1 after day 5
S4	Second miss, not consecutive, inside 7 days	M X M X M	Breaks on day 4; streak is 1 after day 5 (strict reading; open question 2)
S5	Misses 7 days apart	M X M M M M M M X M (misses on days 2 and 9)	Both forgiven; streak is 8
S6	Misses 6 days apart	M X M M M M M X M (misses on days 2 and 8)	Breaks on day 8; streak is 1 after day 9
S7	Today not logged yet	M M, today open	Streak shows 2; no miss until the day ends
S8	Paused days	M P P M	2; pause neither counts nor uses the forgiveness
S9	Midnight boundary	Logs at 23:59 and 00:01 local	Two separate days, in every test time zone
S10	Time zone change mid-streak	Change zone on day 3	Streak unchanged; past `local_date` values are not rewritten
S11	Delete a log within 7 days	M M M M, delete day 3	Streak 3; day 3 becomes a half dot
S12	Property	Any log order, recompute twice	Same result every time

Goals (FR-GOL-04 to 08, FR-TRT-04, 05)

#	Setup and action	Expected
G1	Count goal, target 20; logs reach 10, then 18, then 20, then more	One 50% nudge at 10, one 90% nudge at 18, completed and celebrated at 20, no further nudges

#	Setup and action	Expected
G2	"8 glasses a day for 7 days"	A day with 7 glasses does not count; 8 does; completes on the 8th glass of the 7th day
G3	Completing log is undone within 10 seconds, treat unclaimed	Goal returns to active; no Trophy entry
G4	Deadline passes at 60% progress	Status is due; extend, shrink and restart all work and keep history; never a failed state
G5	Limit goal "under Rs 500 a day for 5 days"	Day totals of Rs 499 and Rs 500 count; Rs 501 does not; today's open day is not counted until it ends
G6	Edit a log from 3 days ago from 8 glasses to 2	Progress recomputes; the same edit on a log 8 days old is refused
G7	Goal on water; goal on Food spending	A gym log does not move the water goal; a Transport expense does not move the Food goal
G8	Recompute twice, shuffle log order, replay after sync	Identical goal rows each time
G9	Goal completed offline; another device deleted one of its logs	Server state is adopted quietly; a claimed treat is not taken back
G10	Claim, then Later, on two completed goals	Claim adds a dated trophy; Later keeps the treat on the Goals tab

When one log crosses a nudge threshold and completes the goal at once, this plan expects only the celebration to show (open question 3).

Safe-to-spend and budget (FR-MON-06, 07)

Amounts are in paise. Safe to spend = (budget minus counted spend) divided by days left including today, rounded down, never below zero.

#	Case	Expected
B1	Budget 3,000,000; spent 0; 1 October (31 days left)	96,774 (Rs 967.74)
B2	Budget 3,000,000; spent 1,000,000; 15 October (17 days left)	117,647 (Rs 1,176.47)
B3	Budget 3,000,000; spent 3,200,000	0
B4	Budget 3,000,000; spent 2,500,000; 31 October (1 day left)	500,000 (Rs 5,000)
B5	Budget 2,800,000; spent 0; 1 February 2027 (28 days)	100,000 (Rs 1,000)
B6	A Rs 600 paid treat is logged	Value unchanged
B7	An income entry is logged	Value unchanged
B8	An expense at 00:30 on 1 November in Asia/Kolkata (19:00 UTC on 31 October)	Counts in November for a Kolkata user, in October for a UTC user
B9	No budget set	No number shown; the app invites setting one
B10	Food limit Rs 5,000: spend Rs 3,999, then Rs 4,000, then more, then Rs 6,000	No warning at 3,999; one calm warning at 4,000; no second warning that month; every entry is still saved
B11	100 paise over 3 days	33; rounding is always down

XP (FR-ENG-04, FR-MND-13)

Awards are 1 per water glass, 20 per workout, 15 per journal entry and 5 per mood check-in, once per log: an edit or a retry never awards twice, deleting a log never removes XP, and a mood of 1 earns the same 5 as a mood of 5. What happens to XP when a log is undone is not specified (open question 4).

6. Offline and sync scenarios

Lost data is the failure users forgive least, so sync gets its own scenarios. The end-to-end ones run in Playwright with the browser's offline switch and a controllable server; the integration ones call the sync routes directly.

#	Scenario	Expected result	Level
Y1	After one online visit, go offline, reload, log a habit, an expense and a mood	The app opens; all three are saved and shown at once; the badge "Saved on this device" appears	E
Y2	Reconnect after Y1	Entries sync with no action within a minute; the badge disappears; the server holds each entry once	E, I
Y3	Close the tab or kill the browser while offline with queued entries	Nothing lost on reopening; the queue still syncs	E
Y4	Start a timer, close the tab, return 10 minutes later and stop	Elapsed time is about 10 minutes more than at close; one log is saved	E
Y5	The same entry is edited on device A and device B, then both sync	The later edit wins on both; the earlier is reported as stale and replaced	I, E
Y6	One device deletes a log while another edits it	Whichever has the later time wins on both devices	I
Y7	A lost response: the server applies a batch but the client sees a failure, then resends	No duplicates; the second send returns "duplicate" for each mutation	I
Y8	Server returns 500, times out, or answers 503	Entries stay queued; retries back off to at most 30 seconds; nothing is deleted locally	I, E
Y9	A batch over 200 mutations or 1 MB	Rejected as too large; the client splits it and every part syncs	I
Y10	A mutation is rejected (for example `log_too_old`)	It is not retried unchanged; local data is kept; the person sees a calm message only if it concerns them	I

#	Scenario	Expected result	Level
Y11	A device clock 10 minutes ahead writes an edit	The server clamps the time to no more than 5 minutes ahead; the edit is still applied	I
Y12	Two changes get sequence numbers that commit out of order	The client's overlap re-pull still receives both; no row is skipped	I
Y13	A device unseen for over 90 days pulls	The server answers `cursor_too_old`; the client does a full resync and ends with the same data as the server	I, E
Y14	A new release raises the minimum schema version while an old tab is open	The old client gets `client_too_old` and is asked to reload; no writes are accepted from it	I, E
Y15	Goal completed offline, then sync	Celebration shown instantly; confirmed after sync; if the server differs it is adopted quietly	E
Y16	Guest merge interrupted halfway, then repeated	End state equals one clean merge; no duplicates	I
Y17	Storage cleared by the browser for a guest	Manual check on real Safari and Chrome: the app starts clean without a crash and the onboarding explains guest data lives on the device	M

Pass rule. Y1 to Y4, Y7 to Y8 and Y16 must pass before the beta release; they are the direct proof of NFR-04 (no log is lost) and NFR-11 (offline). Y17 cannot be fixed by code alone, so its result informs the wording and the sign-in prompt.

7. Non-functional, accessibility and compatibility tests

Every PRD target gets a method, a pass criterion and a time it runs. Rows marked beta must pass before the beta release (NFR-01, 02, 04, 05, 06, 08, 11).

NFR	How it is tested	Pass criterion	When	Beta
01 Speed: Today interactive	Lighthouse CI on a throttled mid-range profile over 4G in CI; the same on a real mid-range Android phone	Under 2 seconds	Every pull request; pre-beta on device	Yes
02 Speed: tap and save	Performance trace of a habit tap and an expense save; "saved" means committed on the device	Card updates within 100 ms; saved within 300 ms	Nightly; pre-beta on device	Yes
03 Availability	Uptime monitor on a health route; incident log	99.5% a month (to validate with beta data)	Continuous	No
04 No log lost	Scenarios Y1 to Y4, Y7, Y8, Y16	All pass	Every pull request (core), nightly (all)	Yes
05 Security	Section 8	No open severe finding	Pre-release	Yes
06 Privacy	Section 8 data audit	No journal text or exact amount outside allowed places	Pre-release	Yes
07 Export and deletion	End-to-end: create data in every table, export, delete, run the purge, check every table	Export is complete; no rows remain; done within 30 days	Pre-V1 and pre-public	No

NFR	How it is tested	Pass criterion	When	Beta
08 WCAG 2.2 AA	Axe on every screen and state in CI; manual keyboard and screen-reader pass	No axe violations; 4.5:1 text contrast, 48 px targets, visible focus, labels on icons	Every pull request; pre-beta	Yes
09 Motion and color	Run with reduced motion on; view in grayscale	Fade replaces confetti and breathing animation; nothing relies on color alone	Pre-beta	No
10 Browsers and width	Playwright projects and manual checks, matrix below	Core flows work in all cells, from 360 px	Nightly; pre-release	No
11 Offline	Airplane-mode test on real devices plus Y1	Opens and logs with no connection after a first visit	Pre-beta	Yes
12 Scale	k6 on staging against the sync routes with synthetic users, ramped toward 50,000 users′ worth of data	Target to validate; record latency and error rate	Pre-public	No
13 Test coverage	CI report on the engine and critical flows	Goal engine, streak and budget code fully covered by automated tests; every critical flow has an end-to-end test	Every pull request	No

NFR	How it is tested	Pass criterion	When	Beta
14 Observability	Trigger a test error and a test event on staging	Error appears in Sentry without personal content; event carries no text or amounts	Pre-beta	No
15 Localisation	Switch currency and locale in a test build	Money and dates format correctly; no hard-coded strings found by a lint rule	Pre-public	No
16 Compliance	DPDP Act 2023 and GDPR review	Legal sign-off	Before public launch	No
17 Content	Copy lint plus human content review	No shaming words, no medical, diet or investment advice, self-care statement visible	Every pull request (lint); pre-beta (review)	Yes (disclaimer)

Compatibility matrix. A cell is checked on the core flows (log, goal and celebration, expense, mood, offline then sync).

Browser	Versions	Where
Chrome	Latest two	Desktop; Android phone, as a tab and installed
Safari	Latest two	iPhone, as a tab and installed to the home screen; macOS
Edge	Latest two	Desktop
Firefox	Latest two	Desktop; Android

Widths tested: 360, 390, 768 and 1280 px, with 200% zoom on the narrowest. Both light and dark themes are checked on every screen.

Manual accessibility pass (before beta). Keyboard-only use of every flow; a screen reader on each platform family (VoiceOver on iPhone, TalkBack on Android, a desktop reader on Windows); reduced-motion on; text enlarged to 200%. The celebration, undo toasts, nudges and the support card are checked specifically, because they appear unprompted.

Real-device checks. A mid-range Android phone (about 4 GB of memory) and an iPhone: install from the browser, airplane mode, receive a reminder by push (iPhone only works when installed) and by email, tap the notification, and confirm it opens Today.

8. Security and privacy tests

Security testing supports NFR-05 and privacy testing supports NFR-06 and NFR-07; both are beta gates. Attack tests always use two accounts, A (the attacker) and B (the victim).

#	Area	Test	Expected
T1	Magic link	Replay a used link; use an expired link; request a link for an unknown email	Used and expired links are refused; the unknown email gets the same response as a known one
T2	Rate limits	Send 6 magic-link requests in 10 minutes for one email; 21 in an hour from one address; 121 writes in a minute	Requests beyond each limit get 429 with `Retry-After`
T3	Sessions	Inspect cookies; use an expired session; sign out then reuse the cookie	HttpOnly, Secure, SameSite=Lax; expired and signed-out sessions get 401
T4	Origin check	Send a write from another origin	403
T5	Access to others' data	A requests B's row ids through sync, export and profile routes	Nothing returned; no error that confirms the id exists
T6	Row-level security	Run a query without the user filter inside a request context	Only the signed-in user's rows come back

#	Area	Test	Expected
T7	Injection	Script text in habit names, notes, tags and treat names; odd characters in ids	Rendered as plain text; the Content Security Policy blocks inline script; validation rejects bad ids
T8	Validation	Unknown fields, negative money, oversized bodies, a fourth mood tag	Rejected with `validation_failed` or 413
T9	Uploads	Treat photo with a wrong type, an oversized file, or a script inside an image	Rejected or re-encoded; never served as a page
T10	Internal routes	Call cron routes with no secret and a wrong secret; call the billing webhook with a bad signature	401 and 400 respectively; nothing runs
T11	Transport	Request the app over plain HTTP; check headers	Redirect to HTTPS; HSTS and a strict Content Security Policy present
T12	Scans	OWASP ZAP baseline on staging; dependency audit; secret scan of the repository and build	No high findings open; no secrets found
T13	Money safety	Search routes, screens, dependencies and permissions for bank linking or payments of user money	None exist (FR-MON-13)

Privacy and data audit

#	Test	Expected
P1	Seed an account with a recognizable journal sentence, a mood note and an odd amount, then use every feature; capture all analytics events, push payloads, emails, Sentry events and server logs	None of them contains the journal sentence, the mood note or the exact amount

#	Test	Expected
P2	Read the journal table directly in the database	Only ciphertext, nonce and key version; the plaintext cannot be found
P3	Export for the account	Every table is present, journal bodies readable, mood and money included, in CSV and JSON
P4	Delete the account	Session ends at once; after the purge job runs, a query across every table holding a user id returns no rows; object storage holds no photos for the user
P5	Backup retention	Check the provider setting and a restore test
P6	Notification copy	Generate every tone for sample users
P7	Support card and disclaimer	Trigger the three-low-day rule and a self-harm phrase in a test journal entry
P8	Analytics allowlist	Try to send an event with a free-text or amount property

The low-mood and self-harm checks (P7) cover P1 requirements FR-MND-08 to 10; they run only if those are built for the launch, but the disclaimer check applies to the MVP in all cases (FR-MND-14). Before public launch, the privacy text, the support-card helpline list and the DPDP and GDPR review each need a named reviewer's sign-off (NFR-16).

9. Pipeline, entry and exit criteria, defect handling

Automation pipeline

Stage	Runs	Gate
Pull request	Typecheck, lint (including copy and hard-coded-string rules), unit and property tests, integration tests on a real Postgres, smoke end-to-end flows in Chromium, axe scan, Lighthouse budget	All green to merge

Stage	Runs	Gate
Merge to main	Same checks, then deploy to staging and run the smoke test there	Green to deploy
Nightly	Full end-to-end suite in Chromium, WebKit and Firefox, all offline and sync scenarios, k6 smoke, ZAP baseline, dependency and secret scans	Failures open a defect; two nights red blocks release
Pre-release	Manual checklists (devices, screen readers, content, privacy), load test, security review, restore test	Sign-off below
After release	Production smoke test, uptime and error dashboards	Rollback if a severe defect appears

Entry and exit criteria by release. These restate the PRD's release criteria as testable conditions.

Release	Must be true to start testing	Must be true to release
MVP beta	All P0 acceptance tests written; builds deploy to staging	All 48 P0 tests pass; NFR-01, 02, 04, 05, 06, 08, 11 pass; no open severe defects; privacy and disclaimer text reviewed
MVP public	Beta gate met: 10 beta users logging 4 or more days a week	P1 requirements chosen for launch pass; DPDP and GDPR review complete (NFR-16); name check done; backups and restore tested
V1	Public MVP stable	The V1 gate (day-30 retention near 20%) is met; all V1 requirements pass; payments tested end to end; export and deletion tested
V2	V1 stable	Day-30 retention and activation targets hold for two consecutive months; the native wrapper passes store review

Defect severity

Severity	Meaning	Examples	Rule
1 Blocker	Data loss, privacy or security exposure, or the app cannot be used	A lost log; journal text in an email; one user sees another's rows	Stop and fix; release blocked
2 Major	Core loop broken with a workaround, or a promise broken	Wrong safe-to-spend; a streak that breaks wrongly; a shaming word; a failed state shown	Fix before release
3 Minor	Works but is wrong or awkward	Misaligned control; slow secondary screen	Fix when scheduled
4 Cosmetic	Appearance only	Spacing, icon weight	Backlog

The PRD says "no open severe defects" for the beta; this plan reads "severe" as severity 1 and 2 (open question 5). Every defect fixed in the engine, sync or privacy code gets a regression test in the same change, so the same bug cannot return.

10. Schedule, risks and open questions

Testing runs alongside the eight MVP build weeks in the Technical Specification, not after them: each area's acceptance tests are written before it is built, and its checks run in CI from the week it lands. The PRD says durations stretch if one person builds part-time.

Week	Build	Tests written and run
1	Repo, CI, schema, auth, guest mode, PWA shell	CI pipeline; constraint tests; auth and rate-limit tests (T1 to T4); FR-ACC-01, 02, 04
2	Local store, engine package, habits, Today	Engine skeleton and S1 to S12; FR-HAB tests; axe and Lighthouse on Today from the first screen
3	Sync, guest merge, offline	Y1 to Y16; FR-ACC-03; FR-SYN tests; first run on a real iPhone and Android phone

Week	Build	Tests written and run
4	Goal engine, treats, celebration, claim	G1 to G10; FR-GOL and FR-TRT tests
5	Streaks, daily ring, expenses, budget	S-cases complete in all time zones; B1 to B11; FR-MON tests; FR-ENG-01
6	Mood, journal, disclaimer, support card if time allows	FR-MND tests; P1, P2, P6; copy lint
7	Reminders, onboarding, settings, export, deletion	FR-NOT and remaining FR-ACC tests; P3, P4, P5; T5 to T11
8	Hardening and beta gate checks	Device Lighthouse and tap timing; screen-reader pass; ZAP and dependency scans; restore test; full beta checklist

Risks to the plan

Risk	Effect	Response
One person builds and tests	Blind spots in both	Write acceptance tests before code; have the beta users and a second reviewer check content, privacy and copy
Offline and time-dependent tests turn flaky	People stop trusting red builds	Controlled clock and stubbed services; a flaky test is quarantined within a day and fixed within a week
Safari and iOS behave differently	Late surprises in install, push and storage	Real iPhone from week 3, not week 8
Date and time-zone bugs	Wrong streaks or budgets near midnight	The four-zone matrix on every date rule
Performance drifts late	Misses NFR-01 and NFR-02	Budgets enforced in CI from week 2; device check in week 8
Ambiguous requirements	Tests assert the wrong behavior	Close the open questions below before the tests for that area are written

Risk	Effect	Response
No real mid-range devices	Speed figures are guesses	At least one mid-range Android phone and one iPhone, or a cloud device service
Free-tier limits distort the load test	NFR-12 result misleads	Run the load test against a staging database sized like production for that day

Open questions that change the tests

1. **Savings goals in the MVP.** FR-GOL-01 (P0) lists six goal types including savings, but savings goals are P2 and V1 in the Functional Requirements. This plan tests five in the MVP screens and all six in the engine.
2. **Second miss inside 7 days.** Does a second, non-consecutive miss inside the window break the streak? This plan assumes yes (cases S4 and S6).
3. **Nudges versus completion, and what progress counts.** For "8 glasses a day for 7 days", is progress measured in days met or in glasses? In days, the 90% point and completion fall on the same log. This plan expects only the celebration in that case.
4. **XP when a log is undone.** XP is never removed (FR-ENG-04), but undo within 10 seconds could be used to farm XP. Decide whether undo revokes the award.
5. **Meaning of "severe".** Assumed severity 1 and 2.
6. **Display of safe to spend.** Paise are exact (Rs 967.74 in case B1); decide whether the app shows whole rupees and which way it rounds.
7. **Low mood.** FR-MND-08 does not define a low-mood score or how much goals soften; the tests need both before they are written.
8. **Meaning of "mid-range phone".** Name the reference device and its network profile for NFR-01.
9. **Availability measurement.** NFR-03 needs a window and a definition of downtime before it can be checked.
10. **Independent review.** The PRD assigns design, engineering and privacy review to the same person; an independent reviewer for privacy and content is recommended before public launch.